

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
- ✓ **DPP, REVISION SHEETS**
- ✓ **100% MONEYBACK TO MERITORIOUS STUDENTS**

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE Mains 2020 Jan Chapter wise Question Bank

Complex Numbers

Q1

If $z = x + iy$ and real part $\left(\frac{z-1}{2z+i}\right) = 1$ then locus of z is

- (1) Straight line with slope 2 (2) Straight line with slope $-\frac{1}{2}$
 (3) circle with diameter $\frac{\sqrt{5}}{2}$ (4) circle with diameter $\frac{1}{2}$

7th Jan Morning

Sol

(3)

$$z = x + iy$$

$$\left(\frac{z-1}{2z+i}\right) = \frac{(x-1)+iy}{2(x+iy)+i} = \frac{(x-1)+iy}{2x+(2y+1)i} \times \frac{2x-(2y+1)i}{2x-(2y+1)i}$$

$$\operatorname{Re}\left(\frac{z-1}{2z+i}\right) = \frac{2x(x-1)+y(2y+1)}{(2x)^2+(2y+1)^2} = 1$$

02

If $z = \left(\frac{3+i\sin\theta}{4-i\cos\theta}\right)$ is purely real and $\theta \in \left(\frac{\pi}{2}, \pi\right)$ then $\arg(\sin\theta + i\cos\theta)$ is -

यदि $z = \left(\frac{3+i\sin\theta}{4-i\cos\theta}\right)$ पूर्णतया वास्तविक है तथा $\theta \in \left(\frac{\pi}{2}, \pi\right)$ तो $\arg(\sin\theta + i\cos\theta)$ का मान होगा-

- (1) $-\tan^{-1}\frac{3}{4}$ (2) $\pi - \tan^{-1}\frac{3}{4}$ (3) $\pi - \tan^{-1}\frac{4}{3}$ (4) $\tan^{-1}\frac{4}{3}$

7th Jan Evening

Sol

(3)

$$z = \frac{(3+i\sin\theta)}{(4-i\cos\theta)} \times \frac{(4+i\cos\theta)}{(4+i\cos\theta)}$$

as चूंकि z is purely real पूर्णतः वास्तविक है $\Rightarrow 3\cos\theta + 4\sin\theta = 0 \Rightarrow \tan\theta = -\frac{3}{4}$

$$\arg(\sin\theta + i\cos\theta) = \pi + \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right) = \pi + \tan^{-1}\left(-\frac{4}{3}\right) = \pi - \tan^{-1}\left(\frac{4}{3}\right)$$

03

Roots of the equation $x^2 + bx + 45 = 0$, $b \in \mathbb{R}$ lie on the curve $|z + 1| = 2\sqrt{10}$, where z is a complex number then

समीकरण $x^2 + bx + 45 = 0$, $b \in \mathbb{R}$ के मूल वक्र $|z + 1| = 2\sqrt{10}$ पर स्थित है, जहाँ z सम्मिश्र संख्या है तब

- (1) $b^2 + b = 12$ (2) $b^2 - b = 30$ (3) $b^2 - b = 36$ (4) $b^2 + b = 30$

8th Jan Morning

Sol

(2)

Let $z = \alpha \pm i\beta$ be roots of the equation

माना $z = \alpha \pm i\beta$ समीकरण के मूल है

So $2\alpha = -b$ and $\alpha^2 + \beta^2 = 45$, $(\alpha + 1)^2 + \beta^2 = 40$

इसलिए $2\alpha = -b$ तथा $\alpha^2 + \beta^2 = 45$, $(\alpha + 1)^2 + \beta^2 = 40$

So $(\alpha + 1)^2 - \alpha^2 = -5$

इसलिए $(\alpha + 1)^2 - \alpha^2 = -5$

$\Rightarrow 2\alpha + 1 = -5 \Rightarrow 2\alpha = -6$

so $b = 6$

इसलिए $b = 6$

hence $b^2 - b = 30$ अतः $b^2 - b = 30$

04

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $a = (1+\alpha) \sum_{k=0}^{100} \alpha^{2k}$, $b = \sum_{k=0}^{100} \alpha^{3k}$. If a and b are roots of quadratic equation then

quadratic equation is

माना कि $\alpha = \frac{-1+i\sqrt{3}}{2}$ तथा $a = (1+\alpha) \sum_{k=0}^{100} \alpha^{2k}$, $b = \sum_{k=0}^{100} \alpha^{3k}$ यदि a तथा b द्विघात समीकरण के मूल है तो द्विघात

समीकरण है-

(1) $x^2 - 102x + 101 = 0$ (2) $x^2 - 101x + 100 = 0$

(3) $x^2 + 101x + 100 = 0$ (4) $x^2 + 102x + 100 = 0$

8th Jan Evening

Sol

(1)

$\alpha = \omega$, $b = 1 + \omega^3 + \omega^6 + \dots = 101$

$a = (1 + \omega) (1 + \omega^2 + \omega^4 + \dots + \omega^{198} + \omega^{200})$

$$= (1 + \omega) \frac{(1 - (\omega^2)^{101})}{1 - \omega^2} = \frac{(1 + \omega)(1 - \omega)}{1 - \omega^2} = 1$$

Equation : समीकरण $x^2 - (101 + 1)x + (101) \times 1 = 0 \Rightarrow x^2 - 102x + 101 = 0$

05

If $\left| \frac{z-i}{z+2i} \right| = 1$, $|z| = \frac{5}{2}$ then value of $|z+3i|$ is

यदि $\left| \frac{z-i}{z+2i} \right| = 1$, $|z| = \frac{5}{2}$ तो $|z+3i|$ का मान है-

- (1) $\frac{7}{2}$ (2) $\sqrt{10}$ (3) $\sqrt{5}$ (4) $\sqrt{3}$

9th Jan Morning

Sol

(1)

$$x^2 + (y-1)^2 = x^2 + (y+2)^2$$

$$-2y + 1 = 4y + 4$$

$$6y = -3 \Rightarrow y = -\frac{1}{2}$$

$$x^2 + y^2 = \frac{25}{4} \Rightarrow x^2 = \frac{24}{4} = 6$$

$$\Rightarrow z = \pm \sqrt{6} - \frac{i}{2}$$

$$|z+3i| = \sqrt{6 + \frac{25}{4}} = \sqrt{\frac{49}{4}}$$

$$|z+3i| = \frac{7}{2}$$

06

z is a complex number such that $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$ then $|z|$ can't be

z एक समिश्र संख्या इस प्रकार है कि $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$ तब $|z|$ नहीं हो सकता

- (1) $\sqrt{7}$ (2) $\sqrt{10}$ (3) $\sqrt{\frac{17}{2}}$ (4) $\sqrt{8}$

9th Jan Evening

Sol

Complex Numbers

JEE Mains 2020 Jan Chapter wise Question Bank

(1)

$$z = x + iy$$

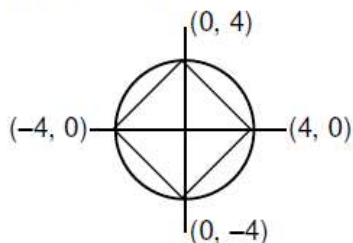
$$|x| + |y| = 4$$

Minimum value of $|z| = 2\sqrt{2}$

Maximum value of $|z| = 4$

$$|z| \in [\sqrt{8}, \sqrt{16}]$$

So $|z|$ can't be $\sqrt{7}$



mathongo

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir

